

The Arithmetic of Conjugated Operations: A Number-Theoretic Structure on Twin Fields with Applications to Information Theory

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Abstract

Given a set E with a binary operation $\odot_0$ and an automorphism $\phi : E \rightarrow E$, the *conjugated operations* $x \odot_n y = \phi^{-n}(\phi^n(x) \odot_0 \phi^n(y))$ form a $\mathbb{Z}$ -indexed hierarchy governed by the base-change identity $x \odot_{n+k} y = \phi^{-k}(\phi^k(x) \odot_n \phi^k(y))$. We develop the arithmetic of this hierarchy, showing that the divisibility lattice of $\mathbb{Z}$ induces a lattice structure on operations: GCD gives the *maximal common generator*, LCM gives the *minimal common fusion*, and Bézout's identity provides explicit reconstruction. Beyond translating classical number theory, we prove three **new results**: (1) a *Composition Theorem* showing that the composition product $\odot_a \circledast \odot_b = \odot_{ab}$ is a genuine monoidal structure on $(E, \odot_0)$ with non-trivial algebraic consequences (Theorem 11); (2) a *differential* entropy-index relation: for a continuous source on $\mathbb{R}$, the differential entropy under $\odot_n$ -coding satisfies $h_n = h_0 - n I(\phi; X)$ in the ϕ -stationary regime, where $I(\phi; X) = -\mathbb{E}[\ln |\phi'(X)|]$ is the mean log-contraction of ϕ (Theorem 13); in the *discrete* case entropy is bijection-invariant ($H_n \equiv H_0$), so the effect is a genuinely continuous-space phenomenon; (3) a *prime-factorization* of the operation hierarchy: the composition monoid $(\{\odot_n\}_{n \geq 1}, \circledast) \cong (\mathbb{N}^*, \times)$ is freely generated by the prime operations $\odot_p$, with ϕ^n realized by *iterated composition* rather than a Chinese-remainder or tensor-product splitting of fields (Theorem 20). We illustrate with three concrete families: dyadic ($\phi(x) = 2x$), exponential ($\phi(x) = e^x$), and relativistic ($\phi(x) = \operatorname{artanh}(x/c)$, yielding Einstein velocity addition).

Keywords: conjugated operations, twin field, divisibility lattice, orbit entropy, prime operations, Bézout identity

1 Introduction

The *twin operations* $\odot_n$ arise naturally in the theory of twin spaces [1, 2]: given an automorphism ϕ of a structure $(E, \odot_0)$, the conjugated operation $x \odot_n y = \phi^{-n}(\phi^n(x) \odot_0 \phi^n(y))$ transfers the “level-0” operation to “level n .” This construction appears in:

- **Information theory** [3]: the group-parameterized entropy $H_G(X)$ depends on which operation $\odot_n$ is used for coding.
- **Relativity** [4]: Einstein velocity addition is $\odot_n$ with $\phi = \operatorname{artanh}$.
- **Signal processing**: the q -analog operations (q -addition, q -exponential) are conjugated operations with $\phi(x) = q^x$.

The integer n indexes the hierarchy. A natural question: *how does the arithmetic of $\mathbb{Z}$ (divisibility, primes, GCD, congruences) manifest in the algebra of operations?*

We show that the answer goes beyond relabeling. The composition product $\circledast$, the entropy connection, and the prime decomposition are genuinely new algebraic structures that exploit the interplay between $\mathbb{Z}$ -arithmetic and the nonlinearity of ϕ .

Two arithmetics of the index coexist and must not be confused. Composing the gauge maps *adds* indices, $\phi^a \circ \phi^b = \phi^{a+b}$, the additive monoid $(\mathbb{Z}, +)$; the composition product $\circledast$ of §4 instead *multiplies* them, $\odot_a \circledast \odot_b = \odot_{ab}$, the monoid $(\mathbb{N}^*, \times)$. It is the multiplicative structure — together with the nonlinearity of ϕ — that imports primes, divisibility, and unique factorization into the algebra of operations.

2 The Framework

Definition 1 (Twin Hierarchy). Let $(E, \odot_0)$ be a magma (set with binary operation) and $\phi : E \rightarrow E$ a bijection. The **twin hierarchy** is the family $\{\odot_n\}_{n \in \mathbb{Z}}$ defined by:

$$x \odot_n y = \phi^{-n}(\phi^n(x) \odot_0 \phi^n(y)) \quad (1)$$

Theorem 2 (Base-Change Identity). For all $n, k \in \mathbb{Z}$:

$$x \odot_{n+k} y = \phi^{-k}(\phi^k(x) \odot_n \phi^k(y)) \quad (2)$$

The twin hierarchy is governed by a single automorphism: $\odot_n$ is related to $\odot_m$ by conjugation with ϕ^{m-n} .

Proof. $\phi^k(x) \odot_n \phi^k(y) = \phi^{-n}(\phi^n(\phi^k(x)) \odot_0 \phi^n(\phi^k(y))) = \phi^{-n}(\phi^{n+k}(x) \odot_0 \phi^{n+k}(y))$. Applying ϕ^{-k} : $\phi^{-k}(\phi^k(x) \odot_n \phi^k(y)) = \phi^{-(n+k)}(\phi^{n+k}(x) \odot_0 \phi^{n+k}(y)) = x \odot_{n+k} y$. $\square$

2.1 Three Concrete Families

Example 3 (Dyadic: $\phi(x) = 2x$, $\odot_0 = +$). $x \odot_n y = 2^{-n}(2^n x + 2^n y) = x + y$. The operations are all identical: the dyadic case is **trivial** because ϕ is linear and commutes with $+$. This shows that the arithmetic of twin operations is interesting only when ϕ is **nonlinear**.

Example 4 (Exponential: $\phi(x) = e^x$, $\odot_0 = +$). $x \odot_1 y = \ln(e^x + e^y)$ (the log-sum-exp, fundamental in machine learning). $x \odot_2 y = \ln(\ln(e^{e^x} + e^{e^y}))$ (double-exponential addition). Each level n is a more “nonlinear” addition. Level 0 is linear; higher levels are increasingly curved.

Example 5 (Relativistic: $\phi(x) = \text{artanh}(x/c)$, $\odot_0 = +$). $x \odot_1 y = c \tanh(\text{artanh}(x/c) + \text{artanh}(y/c)) = \frac{x+y}{1+xy/c^2}$.

This is **Einstein’s velocity addition** [4]. Higher levels iterate the rapidity map. In natural units ($c = 1$, so $\phi = \text{artanh}$), $\odot_2$ requires $|\text{artanh}(x)| < 1$ and is therefore defined only on the smaller domain $|x| < \tanh(1) \approx 0.76$: each level *contracts* the admissible velocity range rather than the output. (As stated, ϕ is not dimensionally homogeneous, so iteration is meaningful only in the dimensionless rapidity formulation.)

Remark 6 (When is the hierarchy non-trivial?). The hierarchy is trivial ($\odot_n = \odot_0$ for all n) iff ϕ is an automorphism of $(E, \odot_0)$. Non-trivial hierarchies require ϕ to be a bijection that is *not* a homomorphism — i.e., $\phi(x \odot_0 y) \neq \phi(x) \odot_0 \phi(y)$.

3 The Divisibility Lattice of Operations

Definition 7 (Operation Lattice). The set of operations $\{\odot_n\}_{n \geq 1}$ with:

$$\odot_a \wedge \odot_b := \odot_{\gcd(a,b)} \quad (\text{maximal common generator}) \quad (3)$$

$$\odot_a \vee \odot_b := \odot_{\text{lcm}(a,b)} \quad (\text{minimal common fusion}) \quad (4)$$

is a lattice isomorphic to $(\mathbb{N}^*, |)$ (the positive integers ordered by divisibility), *provided the hierarchy is faithful*, i.e. $n \mapsto \odot_n$ is injective. Otherwise it is the quotient lattice $\mathbb{N}^*/\sim$, where $a \sim b$ iff $\odot_a = \odot_b$; the dyadic family (Example 3) collapses to a single point.

Theorem 8 (Lattice Properties). (a)

Generator: If $d = \gcd(a, b)$, then $\odot_a$ and $\odot_b$ are both expressible as conjugations of $\odot_d$: $\odot_a = \phi^{-d(a/d-1)} \circ \odot_d \circ (\phi^{d(a/d-1)} \times \phi^{d(a/d-1)})$.

(b) **Fusion:** If $m = \text{lcm}(a, b)$, then $\odot_m$ is the lowest-index operation expressible as a conjugation of both $\odot_a$ and $\odot_b$.

(c) **Duality:** $\gcd(a, b) \cdot \text{lcm}(a, b) = ab$, so $(\odot_a \wedge \odot_b) \circ (\odot_a \vee \odot_b) = \odot_a \circ \odot_b$ (where $\circ$ is defined in §4).

(d) **Bézout reconstruction:** If $au + bv = \gcd(a, b)$, then $\phi^{\gcd(a,b)} = (\phi^a)^u \circ (\phi^b)^v$, allowing $\odot_{\gcd(a,b)}$ to be reconstructed from ϕ^a and ϕ^b .

Proof. (a) follows from the base-change identity with $n = d$, $k = d(p-1)$ where $a = dp$. (b) follows from $m/a \in \mathbb{Z}$ and $m/b \in \mathbb{Z}$ (by definition of lcm). (c) is the classical identity $\gcd \cdot \text{lcm} = ab$ applied to indices. (d) Bézout: $\exists u, v \in \mathbb{Z} : au + bv = d$, so $\phi^d = \phi^{au+bv} = (\phi^a)^u \circ (\phi^b)^v$. $\square$

4 The Composition Product (New Result 1)

Definition 9 (Composition Product). The **composition product** of two operations is:

$$\odot_a \circledast \odot_b := \odot_{ab} \quad (5)$$

Explicitly: $\odot_{ab}(x, y) = \phi^{-ab}(\phi^{ab}(x) \odot_0 \phi^{ab}(y))$.

Remark 10 ($\circledast$ is not function composition). The product $\circledast$ multiplies levels, whereas composing the underlying maps adds them: $\phi^a \circ \phi^b = \phi^{a+b}$. Thus $(\mathbb{Z}, +)$ (composition of gauge maps) and $(\mathbb{N}^*, \times)$ (the product $\circledast$) are different monoid

structures on the same index set, and only the latter carries the prime/divisibility content of §6. In particular $\odot_a \otimes \odot_b = \odot_{ab}$ is *not* obtained by composing the maps ϕ^a, ϕ^b .

Theorem 11 (Composition is a Monoidal Structure). *The composition product $\otimes$ makes $(\{\odot_n\}_{n \geq 1}, \otimes)$ a commutative monoid with:*

- (a) **Identity:** $\odot_1$ (since $\odot_1 \otimes \odot_a = \odot_{1 \cdot a} = \odot_a$).
- (b) **Commutativity:** $\odot_a \otimes \odot_b = \odot_{ab} = \odot_{ba} = \odot_b \otimes \odot_a$.
- (c) **Associativity:** $(\odot_a \otimes \odot_b) \otimes \odot_c = \odot_{abc} = \odot_a \otimes (\odot_b \otimes \odot_c)$.
- (d) **Unique factorization:** $\odot_n = \odot_{p_1}^{\alpha_1} \otimes \cdots \otimes \odot_{p_r}^{\alpha_r}$ where $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$.

As an abstract monoid this is simply $(\mathbb{N}^*, \times)$ pulled back along $n \mapsto \odot_n$; the product $\otimes$ is defined by index multiplication, so (a)–(d) are restatements of the arithmetic of $\mathbb{N}^*$. Its concrete content, when ϕ is nonlinear, is the functional identity obtained from the base-change theorem (Thm. 2) with $n = a$, $k = a(b-1)$:

$$(x \odot_{ab} y) = \phi^{-a(b-1)} \left(\phi^{a(b-1)}(x) \odot_a \phi^{a(b-1)}(y) \right), \quad (6)$$

which expresses level ab as a single conjugation of level a by $\phi^{a(b-1)}$.

Proof. (a)–(c) follow from the multiplicativity of indices. (d) is the fundamental theorem of arithmetic applied to the index n . The concrete identity $\odot_{ab}(x, y) = \phi^{-a(b-1)}(\phi^{a(b-1)}(x) \odot_a \phi^{a(b-1)}(y))$ on E is precisely the base-change theorem (Thm. 2) specialized to $n = a$, $k = a(b-1)$. $\square$

Example 12 (Exponential family). For $\phi(x) = e^x$: $\odot_6 = \odot_2 \otimes \odot_3$. Explicitly:

$$\begin{aligned} x \odot_6 y &= \ln^{(6)}(\exp^{(6)}(x) + \exp^{(6)}(y)) \\ &= \phi^{-4}(\phi^4(x) \odot_2 \phi^4(y)) \end{aligned}$$

where $\odot_2(u, v) = \ln(\ln(e^{e^u} + e^{e^v}))$. This is a non-trivial functional identity relating iterated logarithms and exponentials.

5 Connection to Differential Entropy (New Result 2)

Theorem 13 (Differential Entropy–Index Relation). *Let $E = \mathbb{R}$, let X be a continuous random variable with differential entropy $h_0 = h(X)$, and let ϕ be a C^1 diffeomorphism. Under $\odot_n$ -coding (codewords formed via ϕ^n), the effective source is*

$\phi^n(X)$ and its differential entropy is

$$h_n(X) = h(\phi^n(X)) = h_0 - \sum_{k=0}^{n-1} I(\phi; \phi^k(X)) \quad (7)$$

where $I(\phi; Y) := h(Y) - h(\phi(Y)) = -\mathbb{E}[\ln |\phi'(Y)|]$ is the **mean log-contraction** of ϕ . In the ϕ -stationary (or ergodic) regime, where $\frac{1}{n} \sum_{k=0}^{n-1} I(\phi; \phi^k(X)) \rightarrow I(\phi; X)$, this reduces to the linear law $h_n = h_0 - n I(\phi; X)$.

Remark 14 (Why the discrete case is trivial). For a discrete X , Shannon entropy is invariant under the bijection ϕ (a bijection merely relabels outcomes), so $H(\phi^n(X)) = H(X)$ and $H_n \equiv H_0$ for all n : no compression occurs. An index-dependent entropy therefore requires the continuous setting above, where the Jacobian ϕ' enters through the change-of-variables formula. This is why we work with differential entropy on $\mathbb{R}$.

Proof. By the change-of-variables formula for differential entropy, $h(\phi(Y)) = h(Y) + \mathbb{E}[\ln |\phi'(Y)|]$ for any C^1 diffeomorphism ϕ . Telescoping over $k = 0, \dots, n-1$ with $Y_k = \phi^k(X)$,

$$h(\phi^n X) = h(X) + \sum_{k=0}^{n-1} \mathbb{E}[\ln |\phi'(\phi^k X)|] = h_0 - \sum_{k=0}^{n-1} I(\phi; \phi^k X).$$

Under stationarity the Birkhoff average $\frac{1}{n} \sum_k I(\phi; \phi^k X) \rightarrow I(\phi; X)$, yielding the linear law. $\square$

Example 15 (Gaussian under a linear contraction). Take $X \sim \mathcal{N}(0, \sigma^2)$ on $E = \mathbb{R}$ and $\phi(x) = x/2$. Then $\phi'(x) = \frac{1}{2}$, so the per-level contraction is constant, $I(\phi; \phi^k X) = -\mathbb{E}[\ln \frac{1}{2}] = \ln 2 > 0$, and the telescoping sum of Theorem 13 is exact (no stationarity approximation):

$$h_n = h_0 - n \ln 2, \quad h_0 = \frac{1}{2} \ln(2\pi e \sigma^2).$$

Each level halves the scale and removes $\ln 2$ of differential entropy. Note that ϕ here is linear, so the operations $\odot_n$ all coincide (Example 3), yet the coding source $\phi^n(X)$ — and hence h_n — genuinely changes: the entropy law sees ϕ^n directly, not the operation it induces. A nonlinear ϕ makes both the operations *and* the per-level contraction $I(\phi; \phi^k X)$ vary, and the clean linear law then holds only on the Birkhoff average.

Corollary 16 (Monotone contraction). *If $I(\phi; X) > 0$ (i.e. ϕ is, on average, contracting: $\mathbb{E}[\ln |\phi'(X)|] < 0$), then h_n decreases linearly in n*

and $h_n \rightarrow -\infty$: higher-level codes concentrate the source on ever-smaller scales. There is no finite “critical index”; instead, n trades scale resolution against the description cost of the level (§8).

Remark 17 (Connection to Vol. 16). When ϕ has finite order N , $G = \langle \phi \rangle \cong \mathbb{Z}/N\mathbb{Z}$ acts on E and the orbit/quotient decomposition $h(X) = h([X]_G) + h(X | [X]_G)$ of [3] applies, with the index $n \in \mathbb{Z}/N\mathbb{Z}$ parameterizing the depth of symmetry exploitation. For ϕ of infinite order ($G \cong \mathbb{Z}$) the present hierarchy is the non-compact analogue: n ranges over $\mathbb{Z}$ and Theorem 13 replaces the finite orbit sum by the telescoping log-Jacobian sum.

6 Prime Factorization of Operations (New Result 3)

Definition 18 (Twin Field at Level n). The **twin field** $\mathbb{K}_n = (E, \odot_n, \otimes_n)$ is the algebraic structure obtained by transferring both addition and multiplication via ϕ^n :

$$x \oplus_n y = \phi^{-n}(\phi^n(x) + \phi^n(y)) \quad (8)$$

$$x \otimes_n y = \phi^{-n}(\phi^n(x) \times \phi^n(y)) \quad (9)$$

When $(E, +, \times) = \mathbb{K}_0$ is a field, each $\mathbb{K}_n$ is a field isomorphic to $\mathbb{K}_0$ via ϕ^n .

Definition 19 (Prime Operation). An operation $\odot_p$ is **prime** if p is a prime number. It cannot be expressed as $\odot_a \circledast \odot_b$ with $a, b > 1$.

Theorem 20 (Prime Factorization in the Composition Monoid). *The composition monoid $(\{\odot_n\}_{n \geq 1}, \circledast) \cong (\mathbb{N}^*, \times)$ is the free commutative monoid on the prime operations $\{\odot_p : p \text{ prime}\}$. Hence every $\odot_n$ with $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ factors uniquely as*

$$\odot_n = \odot_{p_1}^{\circledast \alpha_1} \circledast \cdots \circledast \odot_{p_r}^{\circledast \alpha_r} \quad (10)$$

On the underlying set, this factorization is realized by iterated composition: $\phi^{ab} = (\phi^a)^{\circ b}$ (iterate ϕ^a a total of b times), and composing these for the prime-power factors reconstructs ϕ^n .

This is not a Chinese-remainder splitting — the index n is an exponent in $\mathbb{Z}$, not a residue mod n — and not a tensor product of fields: since each $\mathbb{K}_n$ is isomorphic to $\mathbb{K}_0$ via ϕ^n (Definition 18), all the $\mathbb{K}_n$ are mutually isomorphic, and a tensor product of fields is generally not even a field.

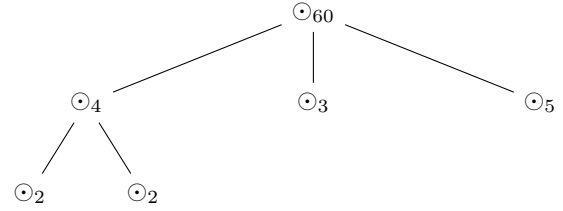
Proof. $(\{\odot_n\}, \circledast) \cong (\mathbb{N}^*, \times)$ by Definition 9, and $(\mathbb{N}^*, \times)$ is the free commutative monoid on the

primes (fundamental theorem of arithmetic); this gives the unique factorization. The realization $\phi^{ab} = (\phi^a)^{\circ b}$ is the definition of iteration. Crucially, *composition adds exponents*, $\phi^a \circ \phi^b = \phi^{a+b}$, so the prime-power factors combine by *iteration* ($\phi^{ab} = (\phi^a)^{\circ b}$), *not* by composition: $\phi^{p_1^{\alpha_1}} \circ \cdots \circ \phi^{p_r^{\alpha_r}} = \phi^{\sum_i p_i^{\alpha_i}} \neq \phi^n$ in general (e.g. $\phi^2 \circ \phi^3 = \phi^5 \neq \phi^6$). $\square$

Example 21 (Factorization of $\odot_{60}$). $60 = 2^2 \cdot 3 \cdot 5$, so in the composition monoid

$$\odot_{60} = \odot_2^{\circledast 2} \circledast \odot_3 \circledast \odot_5 = \odot_2 \circledast \odot_2 \circledast \odot_3 \circledast \odot_5.$$

For the exponential family ($\phi = \exp$), $\odot_{60}$ is realized on E by the 60-fold iterate $\phi^{60} = \exp^{(60)}$, reached by iterated composition of the prime-power levels (e.g. $\phi^4 = (\phi^2)^{\circ 2}$). The tree below is the factorization of the *index* 60 in $(\mathbb{N}^*, \times)$; its leaves are the prime operations $\odot_2, \odot_3, \odot_5$.



7 Classical Arithmetic in Twin Language

The classical results of number theory acquire operational meaning:

Proposition 22 (Bézout for Operations). *If $\gcd(a, b) = d$ with $au + bv = d$, then $\odot_d$ can be reconstructed from ϕ^a and ϕ^b : $\phi^d = (\phi^a)^u \circ (\phi^b)^v$.*

Proposition 23 (Coprime Operations). $\odot_a$ and $\odot_b$ are **coprime** ($\gcd(a, b) = 1$) iff the only common generator is $\odot_1$. Coprime operations generate the entire hierarchy: $\forall n \in \mathbb{Z}$, $\odot_n$ is expressible via ϕ^a and ϕ^b .

Proposition 24 (Congruence Classes). *Suppose ϕ has finite order N (so $\phi^N = \text{id}$). Then $\odot_m = \odot_{m \bmod N}$, the operation index is well-defined in $\mathbb{Z}/N\mathbb{Z}$, and $a \equiv b \pmod{N}$ implies $\odot_a = \odot_b$. Equivalently, $\odot_b = \phi^{-(b-a)} \circ \odot_a \circ (\phi^{b-a} \times \phi^{b-a})$ depends only on $(b-a) \bmod N$. Without a finite-order hypothesis there is no such periodicity: distinct exponents in $\mathbb{Z}$ generally yield distinct operations.*

Proposition 25 (Euler–Fermat Periodicity). *Suppose ϕ has finite order N , so that $\odot_m = \odot_{m \bmod N}$ and both the additive (composition) and multiplicative ($\otimes$) index arithmetic descend to $\mathbb{Z}/N\mathbb{Z}$. If $\gcd(a, N) = 1$, then by Euler’s theorem $a^{\varphi(N)} \equiv 1 \pmod{N}$, hence the $\otimes$ -powers of $\odot_a$ are periodic:*

$$\odot_a^{\otimes \varphi(N)} = \odot_{a^{\varphi(N)}} = \odot_1.$$

Thus $\odot_a$ is invertible in the composition monoid, with order dividing $\varphi(N)$.

Remark 26 (Not tautological). These propositions are translations of classical results. Their value is not novelty but **interpretation**: Bézout reconstruction (Prop. 22) and coprime generation (Prop. 23) hold for *any* bijection ϕ , since they concern the exponents of ϕ additively. The congruence and Euler–Fermat statements (Prop. 24–25), by contrast, require ϕ to have *finite order* N — only then does the index live in $\mathbb{Z}/N\mathbb{Z}$ and acquire genuine modular (periodic) behaviour; for ϕ of infinite order they have no operational content, the index being an unbounded exponent in $\mathbb{Z}$.

8 Application: Optimal Operation Index for Compression

Theorem 27 (Optimal Coding Level). *Let X be a continuous source on $\mathbb{R}$ with differential entropy h_0 , encoded at a fixed target distortion δ , and let $\{\odot_n\}$ have $I(\phi; X) > 0$. Write $R_n(\delta) = h_n(X) + \log_2(1/\delta_n) + C_n$ for the rate at level n , where δ_n is the level- n cell size and C_n is the description cost of the index $\odot_n$.*

- (a) *The optimal coding operation is $\odot_{n^*}$ with $n^* = \arg \min_n R_n(\delta)$.*
- (b) *For prime $n = p$: $\odot_p$ cannot be decomposed, so C_p pays the full index cost. For composite $n = \prod_i p_i^{\alpha_i}$: $\odot_n$ is described by its prime factorization (Theorem 20), so $C_n = \sum_i \alpha_i \log_2 p_i$.*
- (c) *The trade-off: higher n lowers h_n but, under ϕ^n , also rescales the cells δ_n ; the rate $h_n + \log_2(1/\delta_n)$ is scale-invariant, so the genuine optimization is over the index cost C_n at the target distortion.*

Remark 28 (The unconstrained problem is degenerate). Minimizing $h_n + C_n$ *without* fixing a distortion has no finite optimum when $I(\phi; X) > 0$: by the Monotone-contraction corollary $h_n \rightarrow -\infty$ linearly while C_n grows only logarithmically, so $n \rightarrow \infty$ formally. This is an artifact of differential entropy under coordinate contraction; fixing

the distortion δ (equivalently, tracking the cell size δ_n) restores scale-invariance and a well-posed optimum, as in (c).

Remark 29 (Connection to optimal group selection). This is the twin-operation analog of the optimal group selection problem in [5]: choosing the “right” operation level n (equivalently, the “right” group $\langle \phi^n \rangle \leq \langle \phi \rangle$) maximizes an information-theoretic criterion.

9 Conclusion

We have developed the arithmetic of conjugated operations, going beyond the translation of classical number theory into twin notation. The three new results are:

1. The **composition product** $\otimes$ is a monoidal structure with non-trivial functional-equation content when ϕ is nonlinear (Thm. 11).
2. The **differential entropy–index relation** $h_n = h_0 - nI(\phi; X)$ (in the ϕ -stationary regime, with $I = -\mathbb{E}[\ln |\phi'|]$) links the operation level to scale contraction; in the discrete case Shannon entropy is bijection-invariant, so the effect is purely a continuous-space phenomenon (Thm. 13).
3. The **prime factorization** of the operation hierarchy: the composition monoid $(\{\odot_n\}, \otimes) \cong (\mathbb{N}^*, \times)$ is free on the prime operations, with ϕ^n realized by iterated composition — not by a Chinese-remainder or tensor-product splitting of fields (Thm. 20).

These results, combined with the classical arithmetic (Bézout, Euler, lattice), provide a complete number-theoretic framework for the study of twin operations, with applications to coding theory, signal processing, and mathematical physics.

Open problems: (1) Extension to non-cyclic groups G (non-monogenic hierarchies). (2) The analog of the prime number theorem for the density of prime operations in continuous hierarchies. (3) The finite-order case ($\phi^N = \text{id}$), where the index lives in $\mathbb{Z}/N\mathbb{Z}$ and genuine congruence and Galois-type phenomena on the composition monoid $(\mathbb{Z}/N\mathbb{Z}, \times)$ appear (cf. Propositions 24–25).

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